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Influence of metallic contaminants on the electrochemical and thermal behavior of Li-ion electrodes

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ABSTRACT

Emerging nondestructive (direct) recycling techniques for lithium-ion batteries may introduce metallic impurities into recycled electrodes. In the present work, the impact of such nonionic contaminants on the practical performance of both anode and cathode materials is evaluated using a synergistic combination of electrochemical and thermal analysis. The impurities under study have been selected through evaluation of industrially shredded batteries, and include Fe^0, Al^0, Mg^0, Cu^0, and Si^0. The electrochemical behavior of materials containing each individual contaminant at either the anode or the cathode is evaluated in both half-cell and full-cell format. Further, the first-cycle thermal signatures of full cells are used to validate and complement electrochemical signatures, and the two techniques are used in conjunction to suggest distinct mechanisms of electrochemical reactivity for the various impurities. At the anode, metallic contaminants are found to disrupt performance through direct reaction with Li and may serve as weak catalysts to accelerate electrolyte degradation. At the cathode, metallic contaminants show evidence of crossover during formation cycling to disrupt SEI formation. We suggest that coupled electrochemical and thermal analysis may be used to both identify the presence of contaminants and to elucidate specific mechanisms of reactivity for metallic impurities under anodic and cathodic conditions.

1. Introduction

In an increasingly electrified society, lithium-ion batteries (LIBs) have risen to unmatched predominance in the secondary battery market. Growing demand for high-powered personal electronics, electric vehicles (EVs), and grid-scale storage solutions have fueled a nearly $40 billion LIB market – a value that is projected to nearly double by 2026 [1]. By 2028, global production of LIBs is anticipated to exceed 1200 GWh, or about 15 times the LIB capacity produced in 2016 [2]. The number of LIBs reaching their end of functional life will similarly burgeon, offset by the anticipated 8–10 year lifespan of the battery. By 2040, models predict that as many as 2.9 million packs (each containing hundreds to thousands of cells per pack) will be entering the waste stream each year from EVs alone [3]. Developing effective methods to recycle and recover the critical constituent materials of LIBs is crucial to both maintaining material supply chain stability [4] and mitigating environmental hazards associated with disposal.

There are several established approaches to LIB recycling, each relying on a distinct series of processing methods. Pyrometallurgical recycling uses high-temperature smelting to reduce LIB components to their elemental form, which may then be recovered and resynthesized into fresh components. Hydrometallurgical recycling relies on a series of leaching and separation processes to isolate components as salts at the molecular scale. Finally, emerging direct recycling methods seek to recover and refurbish components in nearly intact form, thereby retaining the maximum engineered value. In all reported LIB recycling operations, some form of mechanical tear-down (crushing, shredding) is used to access the internal battery components, reduce particle size, and facilitate downstream treatments [4]. This mechanical treatment, in conjunction with aqueous leaching and separation processes employed in hydrometallurgical recycling, is known to introduce metal impurities into the active materials [5–7]. These impurities may originate from the current collectors (Al, Cu); pack materials (Si, Mg); or the processing machinery itself (Fe).

Several previous studies have investigated these contaminants in the context of a pyrometallurgical or hydrometallurgical recycling scheme, where the metallic impurities are expected to be oxidized to dissolved ions. Unless otherwise removed, these ionic impurities pre-

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sent in the smelted or leached precursors may then be drawn into the synthesis process and substitute into the $Li(Ni_xCo_yMn_{1-x-y})O_2$ (NMC) lattice during resynthesis and sintering. This essentially results in a doping of impurity elements into the NMC, which has been proven to variably impact the performance of the cathode material. Several studies report the benefits of low-level impurity doping on cell performance, with reports of increased capacity retention, thermal stability, and rate performance through lattice stabilization [7–12]. However, inclusion of impurities in cathode resynthesis – particularly in high concentrations – is largely injurious to cell performance, due to morphological impacts on the secondary particle, increased cation mixing, and a reduction in specific capacity [10,11,13–15]. In virgin cathode manufacturing, the extent of doping, substitution, or coatings is an end-user specific choice, and thus the variable presence of impurities in recycled material is typically undesired.

While the study of ionic impurities in recycled NMC is relevant for existing pyrometallurgical and hydrometallurgical processes, these findings cannot be directly translated to instances where contaminants are anticipated to remain in a metallic (nonionic) form. This scenario is anticipated to practically arise in both direct recycling and in the reprocessing of manufacturing scrap. There has been limited work to date investigating the effect of metallic contaminants on battery active material performance. However, the existing literature implies that these impacts may be significant. It has been suggested that larger metallic impurities may physically puncture the separator [16], while smaller metallic impurities may diffuse through the porous separator and migrate to the counter electrode [9,17], leading to internal short circuit in both instances. Inclusion of metallic Al in powder and foil form has been shown to reduce both capacity and Coulombic efficiency of NMC [17], possibly through the surface oxidation of aluminum that impedes Li diffusion. Addition of powdered metallic Cu to NMC has been shown to induce electrochemical failure, ostensibly through the initiation of Li dendrite growth following Cu migration to the anode [9]. There are no existing reports on the impacts of other relevant contaminants (Fe, Mg, Si) in metallic form on cathode performance. Further, the anode has been largely overlooked in existing studies of recycled materials, given the lower material value of graphite as compared to Co and Mn. There are no known reports investigating impurities relevant to recycling and remanufacturing – either ionic or metallic – intentionally introduced into the anode. Finally, the phenomenon of crossover, whereby dissolved transition metals originating from one electrode migrate and are redeposited at the other electrode, has been well-documented [18], and is anticipated to substantially impact the performance of impurity-containing recycled materials.

In assessing the impacts of contaminants on cell performance, *in situ* approaches have generally been limited to electrochemical analysis. Electrochemical techniques enable practical measurement of cell capacities and Coulombic efficiencies, and manipulation of electrochemical data (dQ/dV; CV; EIS) may yield insights regarding physicochemical changes and mechanisms of reaction. However, to glean a complete understanding of what is inherently a chemo-mechanical-electrochemical system, complementary analytical techniques are critical. In previous impurity studies, *in situ* electrochemical data has been coupled with complementary *ex situ* techniques, such as XRD, SEM, and TEM [5]. Only one known study paired *in situ* electrochemical analysis of electrode contaminants with another *in situ* technique. In this report, infrared (IR) thermography was used as a quality control metric to identify regions of electrodes containing metallic defects, which were observed to have reduced emissivity (i.e., more rapid heat dissipation) relative to pristine sites [17]. Such nondestructive, *in situ* analysis of contaminant effects is critical to refining a mechanistic understanding of impurity effects on electrode functionality.

Isothermal microcalorimetry (IMC) is a powerful *in situ* technique by which the thermal response of a cell may be continuously measured during cycling. In the context of intercalation chemistries, IMC has notably been utilized to measure parasitic reactions during cycling [19] [–] [21] and to decouple the Faradic and non-Faradic contributions to cell heating [20–22]. Standard microcalorimetry techniques have previously been applied to the analysis of coin-type cells, which may not accurately represent behavior in larger cell formats (i.e., pouch cells). To that end, we have developed and herein report results from a highly sensitive isothermal microcalorimeter, capable of detecting and quantifying thermal signals as low as 1 mW from pouch cells. Further, our calorimeter design enables deconvolution of heat produced at the anode and at the cathode – a powerful capability for the analysis of practical full cells. Our findings suggest that thermal analysis may be used both to complement electrochemical analysis and as a screening tool to selectively identify impurities in recycled or reprocessed LIB materials.

The present work offers an analysis of the impacts of a variety of metallic contaminants (Al, Cu, Fe, Mg, Si) on both cathode and anode behavior. The choice of these specific impurity elements were based on chemical analysis of industrially generated shredded battery materials (hereafter "black mass"). Using electrochemical methods with microcalorimetric analysis, we highlight the synergy of these two techniques for the detection and specific identification of trace-level contaminants in recycled materials. Further, this coupled approach reveals specific mechanisms of electrochemical reactivity for each contaminant, which may not be evident through each technique in isolation. Taken together, this analysis provides a novel approach to identifying metallic impurities and suggests which metallic contaminants may be most problematic in the context of recycled LIB electrodes.

2. Materials &

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2.1. Industrial black mass: acquisition & characterization

Shredded battery materials (black mass) originating from Li-ion packs and cells was acquired from four industrial suppliers. Processing conditions and shredding techniques vary between suppliers with specifics largely proprietary. Size fractionation via sieving was conducted by Suppliers 1 and 2. Sample designations and processing parameters, where known, are presented in Table 1.

Characterization of the as-received black mass samples included optical imaging and scanning electron microscopy/energy dispersive X-ray spectroscopy (SEM/EDS; Phenom XL Benchtop SEM). Inductively coupled plasma mass spectrometry (ICP-MS) was conducted using a NexION 350D spectrometer (PerkinElmer) with an Opalmist nebulizer. Prior to ICP-MS analysis, NMC samples underwent thermal treatment (to remove residual carbon) and acid digestion. Specifically, triplicate samples were weighed to 100 ± 0.1 g and loaded in an acid-cleaned quartz crucible. Samples were then heated at 700 °C for 2 h in a muffle furnace. Following addition of HCl and HNO_3 (trace metal grade), samples were transferred to an acid-cleaned Teflon microwave reaction vessel, and subject to a microwave reaction program. Once cool, samples were transferred to polypropylene tubes (trace metal grade) and diluted to 50 mL total volume with 18.2 MΩ deionized water. A blank sample

Table 1
Identification of shredded black mass samples obtained from industrial suppliers.

Sample ID	Supplier	Processing Conditions *(Wet vs Dry)*	Shredding Method	Size Fraction
S1	1	Wet	Hammer Mill	>63 μm
S2	1	Wet	Hammer Mill	<63 μm
S3	2	Dry	Shear Shredder	63–350 μm
S4	2	Dry	Shear Shredder	<63 μm
S5	3	Dry	Hammer Mill	–
S6	3	Dry	Hammer Mill	–
S7	4	Dry	Shear Shredder	–

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was subject to the complete digestion and analysis protocol to correct for background and any potential interference.

2.2. Electrode fabrication

All electrodes and cells utilized in this study were produced at the U.S. Department of Energy's (DOE) CAMP (Cell Analysis, Modeling and Prototyping) Facility, Argonne National Laboratory, and were prepared in a dry room operating below −40 °C dew point.

The metallic impurities under study included iron (Fe^0: 10 μm, 99.9% purity, Aldrich); aluminum (Al^0: 5 μm, 99.5% purity, Aldrich); magnesium (Mg^0: 325 mesh (~44 μm), 99% purity, ACROS); copper (Cu^0: 5 μm, 99.8% purity, Goodfellow); and silicon (Si^0: 150 nm, 99.5% purity, Paraclete Energy).

Negative electrodes (anodes) were comprised of graphite powder (Superior Graphite, SLC1520P); carbon black (Timcal C45); binder (Kureha 9300); and oxalic acid (adhesion promotor). Baseline (impurity-free) anodes contained 91.83 wt% graphite, 2 wt% carbon black, 6 wt% binder and 0.17 wt% oxalic acid. Selected impurities were added in metallic form at 1 wt%, substituting for the graphite. Thus, impurity-containing anode compositions were 90.83 wt% graphite, 1 wt% impurity, 2 wt% carbon black, 6 wt% binder, and 0.17 wt% oxalic acid. Metal impurities studied in the anode included Fe^0, Al^0, Mg^0, and Cu^0.

Positive electrodes (cathodes) were comprised of NMC-111 power (Toda, NM-3101); carbon black (Timcal C45); and PVDF binder (Solvay, 5130). Baseline (i.e., impurity-free) cathodes contained 90 wt% NMC-111, 5 wt% carbon black, and 5 wt% binder. Selected impurities were added in metallic form at 1 wt%, substituting for the NMC. Thus, impurity-containing cathode compositions were 89 wt% NMC-111, 1 wt% impurity, 5 wt% carbon black, and 5 wt% binder. Metal impurities studied in the cathode included all of those studied in the anode (Fe^0, Al^0, Mg^0, Cu^0) as well as Si^0.

Mixtures (slurries) with the noted compositions were prepared with a Thinky ARE-310 mixer. For all anodes, the binder solution, carbon black and any metal impurity were first mixed together with two zirconia milling balls (3 min at 2000 rpm followed by a 30 s degas at 2200 rpm; 10 mm mixing ball diameter). The graphite powder was then mixed in, followed by the oxalic acid. A final mixing step was then performed to ensure homogeneity. For all cathodes, the binder solution, carbon black and any metal impurity were first mixed together according to the previously described parameters. Half the amount of the NMC-111 powder and additional NMP was then added, and the solution was remixed. The rest of the cathode powder was added and the solution again remixed, with NMP added and a final mix performed to adjust the viscosity of the slurry.

Anode and cathode slurries were cast using a reverse comma coater (A-Pro; 0.5 m/min; 50 mm coating width). Anodes were coated onto battery-grade Cu foil (LS Mtron; 10 μm) and cathodes were coated onto battery-grade Al foil (Dong-Il Aluminum Co., Ltd; 20 μm). Total anode loading (active material + carbon black + binder) for all electrodes was ~6.4 mg/cm^2 (1.9 mAh/cm^2); total cathode loading was ~11.3 mg/cm^2 (1.6 mAh/cm^2). After coating, all electrodes were roll pressed to a target porosity of 35% using a heated roll press (A-Pro; 80 °C at 0.5 m/min).

2.3. Cell fabrication

The prepared electrodes were assembled into CR2032 coin cells for electrochemical analysis. For half coin cells, an electrode disc (15 mm anode or 14 mm cathode) was paired with a 15.6 mm disc of Li metal (MTI Corporation). Half cells were prepared using a 16 mm disc of Celgard 2320 separator (20 μm thickness) and ~90 μL of Gen2 electrolyte (1.2 M $LiPF_6$ in ethylene carbonate (EC): ethylmethylcarbonate (EMC), where EC:EMC = 3:7 wt%. For full coin cells, a 14 mm cathode was paired with a 15 mm anode. For full-cell testing of cathode contaminants, a baseline (i.e., impurity-free) anode was used as the counter-electrode; for full-cell testing of anode contaminants, a baseline cathode was used as the counter-electrode. Full cells were prepared using a 5/8" diameter disc of Celgard 2320 separator (20 μm thickness) and 29 μL of Gen2 electrolyte. Electrolyte was purchased from Tomiyama Pure Chemical Industries.

Single-layer (SL) xx3450 pouch cells were assembled for thermal analysis using the prepared electrodes and Celgard 2320 separator (20 μm thickness). Cathodes were punched to a coated area of 14 cm^2 (31.44 mm × 45.0 mm) and anodes were punched to a coated area of 14.9 cm^2 (32.4 mm × 46.0 mm), such that the cathode was the limiting electrode and the anode was oversized. Cells were sealed dry (i.e., without electrolyte) under vacuum, with electrolyte added just prior to thermal analysis, as described in Section 2.5.

2.4. Electrochemical cycling

Electrochemical cycling was conducted on half and full coin cells using a multichannel cycler (Maccor). For all cells, formation consisted of 4 cycles of a C/10 constant-current (CC) discharge followed by a C/10 CC charge, with a 2 min rest period between steps. Anode half cells were cycled between 0.005 and 1.5 V; cathode half cells were cycled between 3.0 and 4.3 V; full cells were cycled between 3.0 and 4.2 V. For half cells, long-term cycling (following formation) consisted of 50 cycles at C/3 CC charge and discharge, with a 2 min rest period between steps. Anode half cells were cycled between 0.005 and 1.5 V; cathode half cells were cycled between 3.0 and 4.3 V. For full cells, long-term cycling (following formation) was conducted between 3.0 and 4.2 V. The cycling protocol consisted of a total of 255 cycles, which included five repetitions of the following series of steps: 1) C/20 discharge to 3.0 V, followed by a 10 min rest period; 2) 1 cycle of C/20 CC charge and C/20 discharge, with a 10 min rest period between steps; 2) 48 cycles of C/3 CC charge with a C/5 trickle charge and C/2 discharge; 3) 1 cycle of 1C CC charge (C/5 trickle charge) and 1/C discharge, with a 2 min rest period between steps; 4) 1 cycle of 1C charge (C/5 trickle charge; preparation for Hybrid Pulse Power Characterization (HPPC)); 5) 9 sets of 10-s HPPC pulses, consisting of 5 C discharge followed by 3.75 C charge.

2.5. Thermal analysis

Thermal analysis was conducted using a microcalorimeter designed and fabricated in-house to precisely monitor heat generation from small-format pouch cells. A schematic of the microcalorimeter setup is provided in Fig. A1. Cells were sandwiched between two custom-designed Cu heat spreaders (busbars), with thermal grease applied between the cell and each busbar for enhanced thermal contact. This stack was compressed to 20 psi via calibrated springs. Cell tabs were clamped separately to each busbar to provide electrical contact. Four thermoelectric generators (TEGs; Custom Thermoelectric) were arranged in series on each busbar, with the "hot" side of each TEG oriented towards the battery and the "cold" side oriented outwards. The entire unit was sealed in an Al heat sink, which was then submerged in an isothermal Fluke calibration bath (50 : 50 by volume ethylene glycol: deionized water; stability of ±0.0007 °C at 25 °C). Shielded twisted pair cables, selected to minimize electrical noise, were soldered to the Cu busbars and routed out of the microcalorimeter using liquid-tight pass-through ports.

Prior to analysis, the microcalorimeter was calibrated using a thin Kapton resistive heater, which was installed in the cell site. Six-point calibration was achieved by passing a series of wattages (5 mW–20 mW; Keysight power supply) through the resistive heater, and measuring the TEG voltage output (Keysight Data Acquisition Sys-

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tem 34970A). Linear regression was applied to obtain a calibration curve, which was employed to analyze thermal data (described below).

Immediately prior to thermal analysis, pouch cells produced at the CAMP facility (described in Section 2.3) were opened under Ar atmosphere, filled with 470 μL of Gen2 electrolyte, and re-sealed under Ar. Following installation in the microcalorimeter, the cells were rested for 12 h, which was determined to be the minimum time required to achieve thermal equilibrium. Cells were then subjected to the full-cell formation protocol described in Section 2.4 using a multichannel cycler (Arbin). Electrochemical data was collected via both the Arbin cycler and the Keysight DAQ; thermal data was collected using the Keysight DAQ.

Thermal data was processed using software and analysis techniques developed in-house. Anode and cathode TEG signals were first baseline corrected using the average value measured at thermal equilibrium. A custom smoothing filter ("J-Filter") based on a factored difference between raw signals and an imposed Savitsky Golay filter was applied to mitigate data noise, without compromising peak resolution. In the following report, cell heat output is presented in a voltage-dependent form (W).

3. Results

3.1. Identification of industrially relevant metallic impurities

To develop a system of study with relevance to an industrial recycling/reprocessing line, black mass (shredded battery material) was obtained from four different suppliers (Table 1). Both optical (Fig. A2) and EDS (Fig. 1) images show evidence of impurities in the shredded material. The most common impurity elements reported in literature (Al, Cu, Fe, Si) were observed via EDS; trace levels of P, K, and S were also present in several samples. To quantify impurity quantities, the various black mass samples were analyzed via ICP-MS (Table 2). There is significant heterogeneity in impurity content across suppliers, reflecting both the diversity in shredding and processing techniques utilized and the variety of battery materials analyzed. However, across all samples, Al and Cu – ostensibly from the current collector – are found to be the dominant impurity elements. Fe, Si, and Mg are detected at much lower levels (0.01–1.6 wt %), with measured weight fraction decreasing as Fe ≤ Si < Mg across all samples.

The samples from Suppliers 1 and 2 (S1, S2 from Supplier 1; S3, S4 from Supplier 2) had been size fractionated via sieving prior to acquisition. Samples S1 and S3 contained material from the larger size fraction (>63 μm in each case), while samples S2 and S4 contained material from the smaller size fraction (<63 μm). In both cases, the larger size fraction was found to contain significantly higher levels of Al and Cu

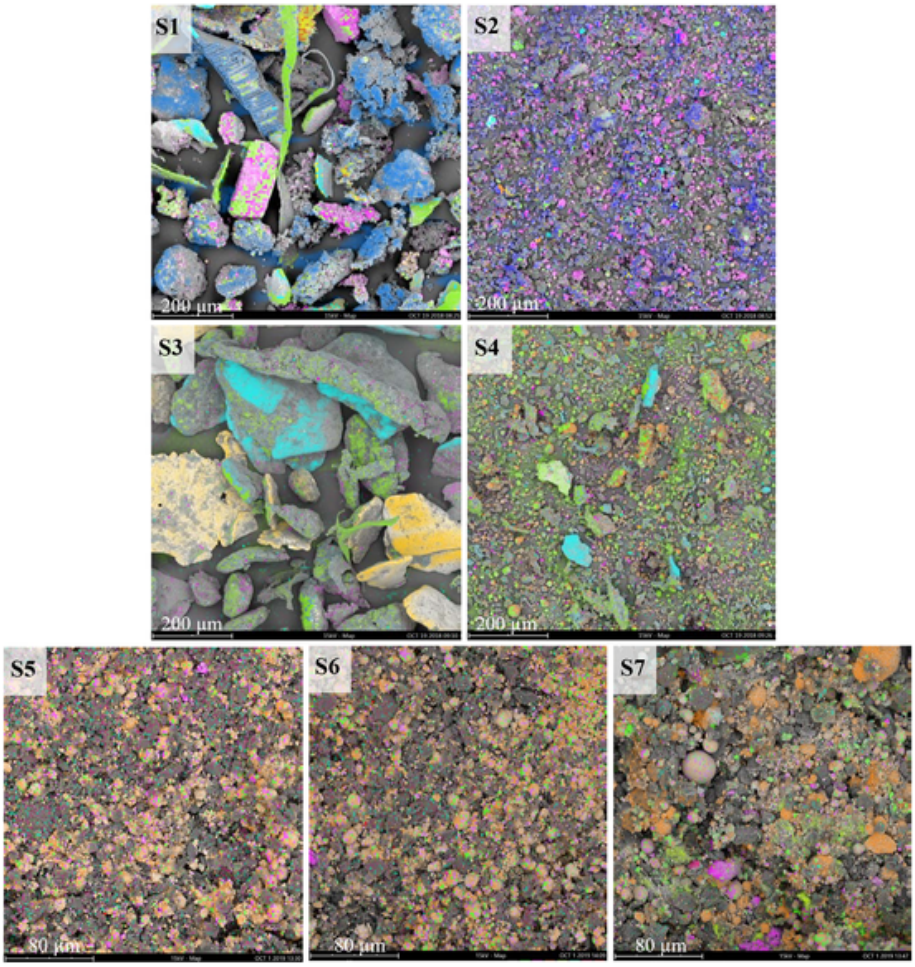

Fig. 1. SEM/EDS data of shredded black mass samples obtained from industrial suppliers. Sample IDs correspond to designations assigned in Table 1. Colors correspond to elements present: **S1**: O = green; Ni, Co = purple; Mn = orange; Al, Si = light blue; Fe, Mg = yellow; P = teal; K = dark blue; **S2:** O = dark blue; Ni = green; Co = purple; Mn = orange; Al, Si = light blue; Fe = yellow; P = teal; **S3–S4:** O = green; Ni, Co = purple; Mn = orange; Al = light blue; Cu = gold; P = teal; **S5–S7:** Ni = green; Co = purple; Mn = orange; Al = yellow; P = teal; S = magenta; Si = red. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

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Table 2
Summary of elemental analysis of industrial black mass samples (ICP-MS). Concentration is reported as a weight percentage of the full sample, with the balance accounted for by standard cathode active material elements (Ni, Mn, Co, O).

Impurity Element[a]	Concentration (wt%), By Sample ID						
	S1	S2	S4	S5	S6	S7	S8
Al	4.3	1.7	16.0	6.6	0.03	0.04	3.07
Cu	1.1	0.6	15.7	3.2	0.55	0.61	1.44
Fe	1.6	0.5	0.4	0.5	0.06	0.07	0.10
Si	1.6	0.4	0.03	0.1	0.02	0.02	0.04
Mg	0.5	0.2	0.01	0.02	0.01	0.01	0.02

[a] Cathode active material elements (Ni, Mn, Co, O) comprise the balance of sample mass.

impurity. For Supplier 1, the larger size fraction contained ~2.5 times the amount of Al impurity and ~1.8 times the amount of Cu impurity as the smaller fraction; for Supplier 2, a similar trend was observed (~2.4 times higher Al and ~4.9 times higher Cu in the larger fraction). Notably, the two suppliers used different mechanical treatment methods: Supplier 1 utilized hammer milling, while Supplier 2 utilized shear shredding. This implies that impurity content in industrial black mass may be more highly dependent on downstream processing (the details of which were proprietary for the materials under study) than on the specific mode of mechanical treatment.

This analysis of industrial black mass informed the design of model systems to investigate the individual impacts of each metallic contaminant on both anode and cathode behavior. In particular, each of the five impurity elements identified in Table 2 (Al, Cu, Fe, Si, and Mg) were studied at a 1 wt% impurity level, which falls within the range of what may practically be anticipated from an industrial recycling or remanufacturing process. Maintaining a constant level of contamination across all samples allows for the direct comparison of specific electrochemical and thermal signatures associated with each impurity. These signatures can then serve to selectively identify the presence of individual contaminants in recycled material.

3.2. *Impact of metallic impurities on anode performance*

The electrochemical signatures of graphite anodes containing four industrially relevant metallic impurities (Fe^0, Al^0, Mg^0, and Cu^0) have been evaluated in both half-cell (CR2032 coin cells; graphite vs Li) and full-cell (pouch cell; graphite vs NMC-111 cathode) formats to assess the impact of each metallic impurity on anode performance. Thermal signatures of contaminated anodes have also been evaluated in full cells. Half cells enable isolation and identification of processes occurring at the anode, which informs a mechanistic understanding of the electrochemical reactivity induced by the presence of contaminants. Full cell testing evaluates the effect of anode contamination in a practical cell format, which is highly relevant in the context of directly recycled materials.

The impacts of metallic contaminants may be most strongly observed during the initial formation cycles, although several contaminant chemistries continue to impact electrochemical performance throughout extended cycling (Figs. A3-A4). Thus, in the present report, emphasis will be placed on analysis of electrochemical and thermal signatures during formation cycling. Specific impacts of each metallic contaminant on the electrochemical and thermal performance of the anode are highlighted, with the aim of both offering a fundamental mechanistic explanation of reactivity and identifying diagnostic features that may be employed to screen for impurities in recycled materials.

Differential capacity plots (dQ/dV) of half cells with impurity-containing anodes, as compared to pristine anodes, are shown for the first four formation cycles in Fig. 2; analogous dQ/dV plots for full cells with impurity-containing anodes are shown for formation cycles in Fig. 3. Differential capacity analysis in full cells results in a convolution of anode and cathode behaviors, which must be decoupled to identify contributions associated with each electrode. Previous reports have undertaken this decoupling via curve-fitting algorithms [23,24], suggesting typical peak positions that may be linked to anodic or cathodic behavior. The present study will utilize these interpretations, as well as comparison with half-cell differential capacity plots, to identify electrochemical changes attributable to each electrode contaminant. It should be noted that the electrochemical performance of cells containing anode impurities varied significantly between cell replicates, perhaps suggesting inhomogeneity in the dispersion of metallic contaminants throughout the electrode. Where relevant, two dQ/dV plots have been included for each cell condition (reflecting two replicates) to more accurately reflect the scope of observed behavior.

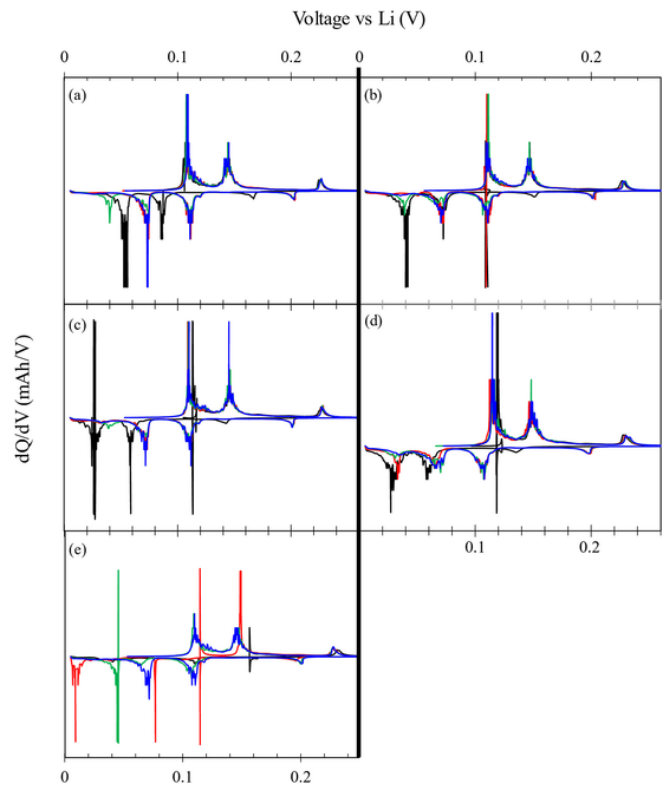

Fig. 2. Differential capacity (dQ/dV) during formation cycles for anode half cells (graphite anode vs Li_0) prepared with metallic impurities: (a) baseline (no impurity); (b) Fe impurity; (c) Al impurity; (d) Mg impurity; (e) Cu impurity. For all plots, cycle 1 = black; cycle 2 = red; cycle 3 = green; cycle 4 = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Finally, thermal signatures for full cells containing metallic anode impurities have been acquired. In Fig. 4, the thermal signatures have been normalized by the signature of a baseline cell so that behavior uniquely attributable to each impurity may be differentially distinguished. Raw thermal signals for each cell are given in Fig. A5.

Each metallic anode impurity will be taken up in detail in the following sections.

3.2.1. *Fe^0-contaminated anode*

The presence of metallic Fe^0 at the anode is generally found to have minimal electrochemical impact. As shown in Fig. 2(b), a slight shift in first-cycle reductive behavior is observed in half-cell format: Specifically, reductive peaks associated with the Fe^0-containing anode are found to shift to lower voltages as compared to the baseline cell, implying reduced thermodynamic favorability. The reductive peak occurring at 0.167 V in the baseline cell is shifted to 0.151 V in the Fe-

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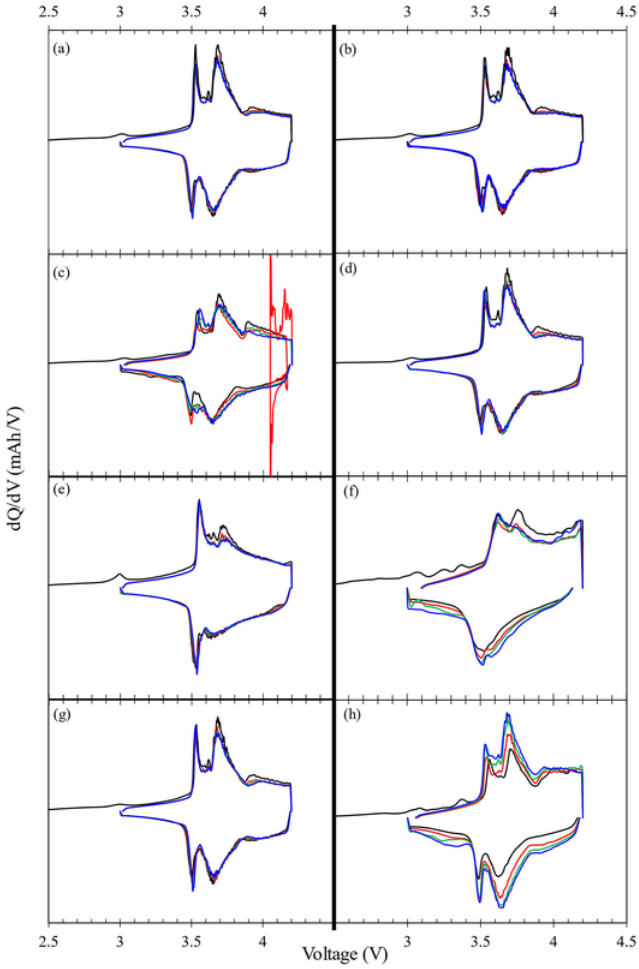

Fig. 3. Differential capacity (dQ/dV) during formation cycles for full cells containing metallic anode impurities: (a) baseline (no impurity); (b) Fe impurity; (c)–(d) Al impurity; (e)–(f) Mg impurity; (g)–(h) Cu impurity. For all plots, cycle 1 = black; cycle 2 = red; cycle 3 = green; cycle 4 = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

contaminated cell; the peak at 0.087 V in the baseline is shifted to 0.072 V; and the peak at 0.056 V in the baseline cell is shifted to 0.042 V. This behavior is limited to the first cycle; at cycle 2 and beyond, the Fe^0-containing anode shows identical electrochemical performance to the baseline cell. Thus, we anticipate that these peaks are associated with irregular SEI formation, likely including the reductive passivation of Fe (e.g., FeO → Fe^0), the dissolution of Li^+ into Fe-containing surface oxides (e.g., $Li_xFe_2O_3$) or weakly catalytic interactions between oxidized Fe^{2+} and electrolyte species. These first cycle losses may contribute to the slight decrease observed in half-cell capacity for Fe^0-containing anodes, relative to pristine anodes (Fig. A3(a)).

In full cells, differential capacity analysis reveals no discernible electrochemical impact on electrochemical performance, as seen in Fig. 3 (b). However, the full-cell specific capacity of cells containing Fe^0 impurity at the anode is reduced relative to the baseline cells, even when the mass of contaminant is omitted from the normalization (Fig. A4(a)). This may be attributed to the irregular first-cycle behavior observed in half cells, which may consume Li^+ through irreversible surface oxide formation. Subsequent surface passivation and the stability of metallic Fe^0 at anodic potentials minimizes further capacity degradation with repeated cycling.

The thermal behavior of full cells containing Fe^0 at the anode is shown in Fig. 4(a). Normalized signal values less than 1 at low potentials (<2.5 V) are attributed to bias error, given the rapid increase of voltage over a brief period of time at the beginning of charge. Thus, the first exothermic signal (~2.25–2.4 V) likely results from Fe^0 oxidation to Fe_2O_3 and Fe_3O_4, both of which occur at low potentials (0.8 V vs Li^+/Li^0) [25]. The observed delay in thermal signal detection is inherent to the kinetics of the process (i.e., heat transfer through the cell, which occurs on the same timescale as the voltage sweep). This error is confined to the initial period of rapid charge, and thus does not propagate. The exothermic peak at ~2.8–3.0 V corresponds to the first-cycle oxidative peak at observed at the same potential in differential capacity analysis (Fig. 3(b)). This electrochemical signal, which is associated with EC reduction to form the SEI [26], is also seen in the first cycle for the baseline cell. However, the fact that a differential exothermic signal is observed *above* the baseline signal for the Fe^0-contaminated cell confirms the participation of Fe^{2+} in irregular SEI formation, and in particular, may suggest a weakly catalytic role of the contaminant in electrolyte reduction. Above 3 V, no difference in thermal signature is observed between the baseline cell and the cell with Fe^0-containing anode.

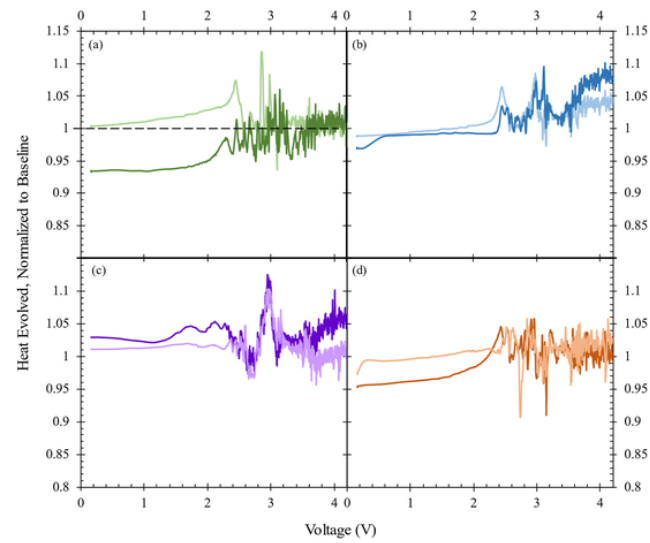

Fig. 4. Thermal signal evolved during first formation charge cycle for full cells containing metallic anode impurities: (a) Fe impurity; (b) Al impurity; (c) Mg impurity; (d) Cu impurity. For each contaminated cell, thermal signal has been normalized to the signature of a representative baseline cell – such that the baseline cell has a normalized value of 1 across all voltages (i.e., the normalized baseline thermal signal can be represented as the dashed black line in (a)). For graphical clarity, the normalized baseline signal has been omitted from plots (b)–(d).

3.2.2. Al^0-contaminated anode

Electrochemical reactivity of a metallic Al^0 impurity with the anode is observed in both half-cell and full-cell format, consistent with the oxophilic [27] and Li-reactive behavior of Al^0 at low potentials. As shown in Fig. 2(c), new dominant oxidative and reductive peaks emerge at 0.027 V during the first cycle; this may be attributable to the evolution of Al–Li Zintl phases, which are anticipated to form at ~400 mV. Notably, this phase formation is only found to occur during the first formation cycle, implying either full consumption of Al^0 impurity with electronic isolation (i.e. trapped in SEI layer) or a passivation of the Al^0 surface, inhibiting further reactivity. Additionally, the first-cycle reductive peak present at 0.167 V in the baseline cell shifts to 0.143 V in the Al^0-containing system as compared to the baseline. The observed downward shift in reduction potential supports the notion of participation of precipitated Al species in the SEI, which may impede intercalation at surface sites.

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The impact of Al^0 impurity is also manifest in full-cell format for a subset of the cell replicates tested. The noise observed at high potential in Fig. 3(c) during the second formation cycle may reflect electrolyte reactivity, and more specifically, increased reactivity associated with the presence of the Al contaminant. Reduced Al species will react with solvents analogously to other main group elements (e.g., Si), and therefore additional reactivity associated with electrolyte decomposition and SEI formation is anticipated for Al^0-contaminated cells. Additionally, upon discharge, a small peak at 3.54 V emerges in the cell with Al^0 impurity. Based on previously-reported deconvolutions [23], it is probable that this peak is associated with the anode, and is masked in the baseline cell as a weak shoulder in the dominant reductive peak at 3.66 V. In the cell with Al^0-containing anode, this reductive peak is found to shift to a lower potential, again implying the disruptive nature of Al^0 on anode surface stabilization.

Notably, a second full-cell replicate containing an Al^0-contaminated anode showed no discernible electrochemical differences as compared to the baseline (Fig. 3(d)), suggesting a batch-scale heterogeneity in contaminant dispersion. Such heterogeneity is anticipated in industrial recycled materials, and thus capturing cell-to-cell variation is critical to accurately representing the impacts of contaminants on recycled material outputs.

As with the Fe^0-contaminated anodes, the Al^0-contaminated anodes generally show a slight decline in specific capacity relative to the baseline material, in both half cell (Fig. S3(b)) and full cell (Fig. A4(b)) format. This suggests irreversible Li^+ loss, which may be attributable to irreversible first-cycle Li–Al Zintl phase formation, side reactions that consume lithium cations, or subtle changes to the SEI composition induced by the aluminum.

The thermal signature of Al^0-contaminated full cell, shown in Fig. 4 (b), is comparable to that of Fe^0-contaminated cells. Notable exothermic peaks occur at ~2.4 V, attributable to Al^0 oxidation (occurring at 1.38 V [28], but delayed in detection due to inherent voltage bias, as discussed in Section 3.2.1) and ~2.8 V, suggesting participation of the oxidized Al species in the EC reduction leading to SEI formation. Notably, the dual peaks at ~2.8 V and ~3.1 V may correspond to reaction associated with the two predominant oxidized forms of Al, namely, Al^{2+} (AlO) and Al^{3+} (Al_2O_3). High-potential thermal signatures correspond neatly to differential electrochemical behavior in this voltage range (Fig. 3(c)), suggesting more highly exothermic electrolyte reactivity in the Al^0-contaminated cells as compared to the baseline cells. It is noted that, although Al^0 is present in the baseline cell (as cathode current collector), the current collector foil is ostensibly pre-passivated with a stabilizing layer of Al_2O_3 due to atmospheric oxidation. Thus, the inclusion of practical contamination from pre-passivated current collectors – rather than pure Al^0 powder – may alter the observed thermal signature; this should be evaluated in future work.

3.2.3. Mg^0-contaminated anode

Half-cell analysis of Mg^0-containing anodes, given in Fig. 2(d), suggests phase formation behavior at low potential (~0.028 V), similar to that observed for anodes with Al^0 impurity. However, in contrast to Al^0-contaminated anodes, the Mg^0-contaminated anodes show continuous reactivity throughout the four formation cycles. Even in the presence of an infinite Li reservoir (i.e., half cell format), such Mg–Li alloying appears to significantly impact cell capacity (Fig. A3(c)), ostensibly through reduced ionic diffusion at the metallic Li surface. The impact on full cell capacity (Fig. A4(c)) is heterogeneous by replicate, with two of the four cell replicates showing drastically reduced capacity. This is consistent with an irreversible loss of cyclable Li^+ due to alloying reactions with Mg.

Half cells containing Mg^0 impurities also show increased hysteresis in the dominant redox peak following the first formation cycle. In particular, the oxidative peak is found to shift from 0.109 V in the baseline cell to 0.120 V in the cell with Mg^0-contaminated anode, with no evident change in the position of the reductive peak. Increased hysteresis indicates increasing electrode impedance, which could be linked to the formation of diffusion-inhibiting Li–Mg alloys. This may also alter the practical potential experienced by the cathode. Finally, the first-cycle reductive peak at 0.167 V in the pristine material shifts to 0.136 V in the Mg^0-contaminated material, suggesting a more thermodynamically unfavorable passivation.

In full cells, a significant decline in intensity of the dominant redox peak at 3.65 V is observed for all cell replicates (Fig. 3(e)–(f)), particularly upon discharge (reduction). This redox peak is typically associated with cathodic behavior [23], supporting the notion that the increased hysteresis induced by the Mg^0 contaminant may be impacting the true cathodic potential. Additionally, a new peak is observed at high potentials (~4.2 V), suggesting increased electrolyte reactivity in the presence of oxidized Mg species.

Slight variability in thermal signature is observed between replicates of Mg^0-containing cells. Specifically, one cell shows low-potential exothermic behavior (ostensibly Mg^0 oxidation to MgO), while the other does not; the former cell also shows increased exothermic signal at high potentials. This cell-to-cell heterogeneity may be associated with local differences in contaminant size and loading, and reflects the range of behaviors that may be observed in cells fabricated from a true end-of-life black mass. The predominant thermal peak for both cells occurs at ~2.75 V, and is likely tied to EC reduction – implying the accelerating impact of Mg (and likely Mg–Li alloy) on electrolyte decomposition and associated SEI formation.

3.2.4. Cu^0-contaminated anode

The behavior of anodes containing Cu^0 impurities is highly erratic during formation cycling in half-cell format, as shown in Fig. 2(e), with dQ/dV peak position varying significantly between cycles. The first formation cycle shows behavior similar to the baseline material; this is anticipated, as Cu is not expected to react with Li at anodic potentials. However, in the subsequent formation cycles, several erratic spikes in differential capacity are observed, implying that the metallic Cu^0 impurity is not entirely inert.

In full cells, the impact of Cu^0 contamination at the anode is heterogeneous between replicates. Some cells show no observable electrochemical difference from the baseline (Fig. 3(g)), but others show a trend of *increasing* dQ/dV intensity during formation (i.e., from cycle 1 to cycle 4; Fig. 3(h)). This trend is also manifest in full-cell capacity analysis (Fig. A4(d)), where two of the four replicates are found to show increased capacity relative to the baseline after extended cycling. We suggest that Cu impurities may act as a conductive additive when sited at the anode, increasing electrode conductivity and thereby enhancing cell performance. However, given that this effect is not observed uniformly between cell replicates, further study should be undertaken to further probe the source of this behavior.

Finally, increased reactivity at high potentials (~4.2 V) is observed for Cu^0-containing anodes in full cells, similar to that seen for Mg_0-contaminated anodes. This may be due to disruption of surface sites by oxidized Cu species (CuO, Cu_2O), or enhanced reactivity induced by ionic Cu at high potential.

Minimal to no impact on thermal signature is observed for the Cu^0-containing full cells, as compared to the baseline cell. This is physically reasonable: Given that Cu is already present in the baseline cell (as current collector), reactivity from Cu oxidation would be captured in the baseline signature; the lack of an additional peak at ~2.8 V, as has been observed for the other anode contaminants, implies a minimal impact of Cu on SEI formation.

3.2.5. Summary of anode contaminant behavior

From a cell performance perspective, Cu^0 and Mg^0 are found to be the most disruptive anode contaminants in both half-cell and full-cell format. Predominant modes of reactivity observed for anode contami-

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nants include alloying and Zintl phase formation at low potentials, and catalytic acceleration of electrolyte decomposition induced by the high oxophilicity of the contaminants under study. The observed reduction in long-term cycling capacity, particularly for cells with Cu^0 and Mg^0 at the anode, is attributed to an increase in the irreversible consumption of Li during initial formation cycling, which appears to form an inefficient and resistive SEI at the anode that inhibits subsequent cycling performance. For all anode contaminants except Cu^0, analysis of thermal reveals the role of the oxophilic impurities in enhancing electrolyte degradation, which is not immediately evident through electrochemical analysis alone. Finally, there is no evidence of physical penetration of the separator (i.e., hard internal short circuit) for any of the metallic contaminants.

3.3. Impact of metallic impurities on cathode performance

As for the contaminated anodes described in **Section 3.2.**, the electrochemical and thermal signatures of contaminated cathodes has been evaluated using a combination of half-cell and full-cell analysis. The cathode impurities studied include Fe, Al, Mg, Cu, and Si. The disparity in anodic and cathodic operating potential leads to differing impacts on cell performance when these contaminants are sited in the cathode, rather than in the anode, and such distinction is crucial to understanding the range of impacts associated with metallic contaminants introduced during recycling.

Differential capacity plots (dQ/dV) of half cells with impurity-containing cathodes, as compared to pristine cathodes, are shown for the first four formation cycles in Fig. 5; analogous dQ/dV plots for full cells with impurity-containing cathodes are shown for formation cycles in Fig. 6. The electrochemical performance of cells containing cathode impurities was more consistent between cell replicates than the performance of cells containing anode impurities, and thus, a single representative dQ/dV plot is shown for each cell condition. While emphasis will again be placed on electrochemical signatures during formation, the capacities of half and full cells containing cathode contaminants over the course of extended cycling are provided in the Appendix (Figs. A6-A7).

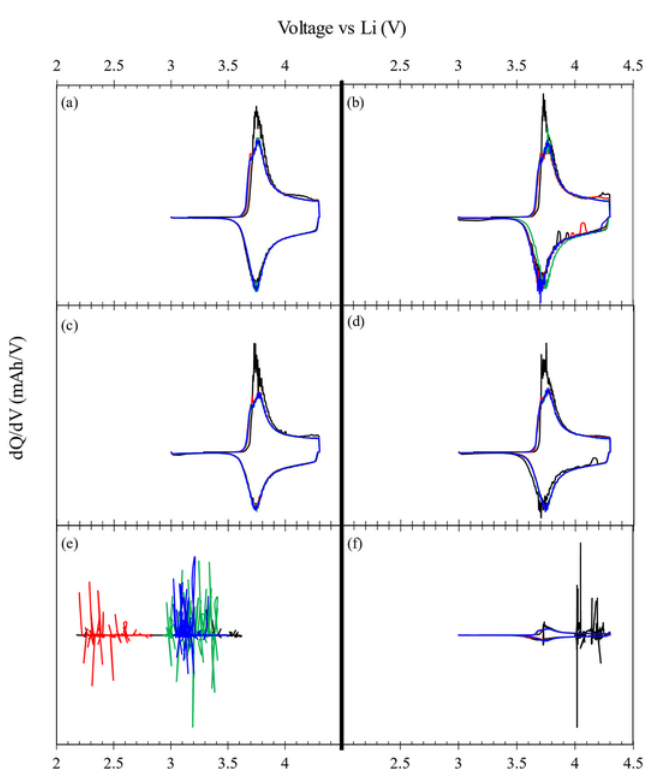

Fig. 5. Differential capacity (dQ/dV) during formation cycles for cathode half cells (NMC-111 vs Li^0) prepared with metallic impurities: (a) baseline (no impurity); (b) Fe impurity; (c) Al impurity; (d) Mg impurity; (e) Cu impurity; (f) Si impurity. For all plots, cycle 1 = black; cycle 2 = red; cycle 3 = green; cycle 4 = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

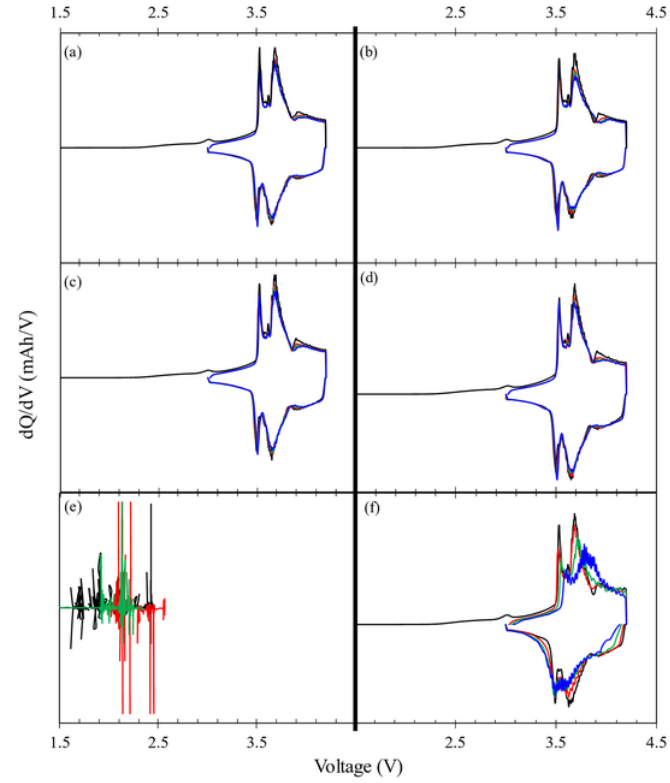

Fig. 6. Differential capacity (dQ/dV) during formation cycles for full cells containing metallic cathode impurities: (a) baseline (no impurity); (b) Fe impurity; (c) Al impurity; (d) Mg impurity; (e) Cu impurity; (f) Si impurity. For all plots, cycle 1 = black; cycle 2 = red; cycle 3 = green; cycle 4 = blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Thermal signals for full cells with metallic impurities at the cathode are presented in Fig. 7. As in Fig. 4, the signals have been normalized by the signature of a baseline cell to highlight the distinguishing features attributable to the impurity. Raw thermal signals for each cell are given in Fig. S8.

Each metallic cathode impurity will be discussed in the following sections.

3.3.1. Fe^0-contaminated cathode

Fe is anticipated to be electrochemically and chemically stable in the cathodic voltage range, and thus a minimal impact on electrochemical performance would be expected. In full-cell format, no noticeable change in electrochemical signature is observed for the Fe^0-contaminated cathode, as compared to the baseline material (Fig. 6(b)). However, in half cells, differential capacity plots show irregularity during the highest voltage portion of charging (oxidation) and the high-voltage portion of discharge (reduction) during the two initial formation cycles (Fig. 5(b)). Interestingly, this corresponds to the voltage regime in which the most significant heat is evolved from Fe^0-contaminated full cells, as shown in Fig. 7(a). This implies a weakly catalytic role of the oxidized Fe^0 impurity in electrolyte decomposition, which occurs at high potentials. The fact that such reactivity in full cells is observable via thermal analysis, but not via electrochemical analysis,

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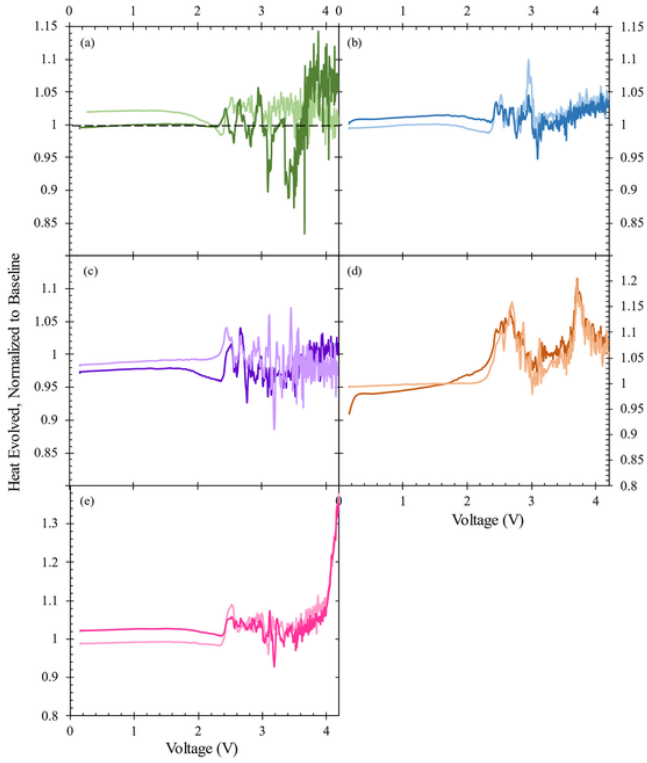

Fig. 7. Thermal signal evolved during first formation charge cycle for full cells containing metallic cathode impurities: (a) Fe impurity; (b) Al impurity; (c) Mg impurity; (d) Cu impurity; (e) Mg impurity. For each contaminated cell, thermal signal has been normalized to the signature of a representative baseline cell – such that the baseline cell has a normalized value of 1 across all voltages (i.e., the normalized baseline thermal signal can be represented as the dashed black line in (a)). For graphical clarity, the normalized baseline signal has been omitted from plots (b)–(e).

highlights the complementary role of these two techniques in identifying the presence of metallic contaminants.

A more significant capacity decline is observed for full cells containing Fe^0 contaminant in the cathode than for half cells with the Fe^0 impurity (Fig. A6(b), Fig. A7(b)). This observed decrease in capacity stands in contrast to the evident lack of reactivity suggested by differential capacity analysis, and supports the notion of Fe^0 playing a catalytic role in electrolyte decomposition pathways and increasing irreversible Li^+ consumption during the initial formation cycles.

3.3.2. Al^0-contaminated cathode

In half cell format, cells containing Al^0 impurity show novel oxidative peaks at 3.96 V and 4.0 V during the first formation cycle (Fig. 5(c)); there is also a slight first-cycle oxidative peak increase at the highest potentials (~4.2–4.3 V). While the formation of Al_2O_3 is theoretically anticipated to occur at relatively low potentials (~1.39 V vs Li^+/Li [29]), we suggest that the oxygen required for Al oxidation may be evolved during electrolyte decomposition, which does not occur until higher potentials. Further, it has been reported that the Al_2O_3 may subsequently react with the fluorinated electrolyte salt ($LiPF_6$) to fluorinate the aluminum surface layer [30]. We suggest that the first novel oxidative peak (3.96) may be associated with Al_2O_3 passivation once the electrolyte has decomposed to produce free oxygen, while the second oxidative peak (4.0 V) may be associated with the fluorination of the Al_2O_3 passivation layer through reaction with HF (produced from decomposition of the alkali electrolyte salts) [30]. Following this first-cycle passivation, the Al^0 contaminant appears to be electrochemically inert.

Minimal to no impact is observed from cathode-sited Al^0 impurity on the full cell differential capacity signature (Fig. 6(c)), and the capacity of full cells containing Al^0-contaminated cathodes is on par with the baseline cells during extended cycling (Fig. A7(c)). The thermal signatures for cells contaminated with Al^0 at the cathode show a spike in heat evolved at ~3.0 V, followed by a gradual rise in heat generation at higher potentials (Fig. 7(b)). It is notable that this thermal signal evolves above the signal of the baseline cell, suggesting specific reactivity of the Al^0 contaminant, rather than the cathode current collector. This signal does not directly align with features observed in the electrochemistry, and thus the mechanism of reactivity leading to this heat generation is not immediately apparent. Taken together, the observed thermal and electrochemical behavior for Al^0-contaminated cathodes implies Al^0 reactivity under cathodic conditions, but with a minimal impact on practical full-cell electrochemical performance.

3.3.3. Mg_0-contaminated cathode

As for Al^0 contaminants, Mg^0 impurities show minimal impact on half-cell cathode performance (Fig. 5(d)), and no distinguishable impact on full-cell electrochemical signature (Fig. 6(d)). However, there are critical differences in the half-cell differential capacity plots that indicate a varying mode of reactivity between these two contaminants. In particular, as seen in Fig. 5(d), Mg^0-contaminated cathodes show irregular reductive behavior at 4.16 V during the first cycle, and a shift to lower voltage in the first-cycle dominant reductive peak, from 3.78 V in the baseline material to 3.70 V in the Mg^0-containing materials, is observed. As has been suggested for Mg^0 contamination at the anode, it is probable that the Mg^0 contaminant may be impacting the practically manifest cathodic potential, leading to the observed increase in hysteresis of the dominant redox peak. Additionally, a new peak is observed at high potentials (~4.2 V), suggesting increased electrolyte reactivity in the presence of oxidized Mg species.

Given the moderately high oxophilicity of Mg^0, it is probable that a stable and passivating film of MgO_2 may form on the Mg^0 surface, with oxygen evolved from electrolyte reduction. This leads to the complete passivation of the Mg^0 contaminant after the first cycle, resulting in minimal disruption of electrochemical performance. The lack of practical impact of the Mg^0 contaminant is evident in both the thermal signature (Fig. 7(c)) and electrochemical capacity (Fig. A7(c)) of full cells containing Mg^0-contaminated cathodes.

3.3.4. Cu_0-contaminated cathode

At cathodic potentials, metallic Cu^0 is highly unstable; this is manifest as a significant disruption in the electrochemical signatures of both half cells (Fig. 5(e)) and full cells (Fig. 6(e)). At potentials around 2.8 V, Cu^0 will oxidize to Cu^+, which is soluble in electrolyte; at higher potentials (around 3.7 V), Cu^+ will further oxidize to soluble Cu^{2+}. These two oxidation potentials are clearly observed in the thermal profiles of full cells contaminated with Cu^0 at the cathode: As shown in Fig. 7(d), exothermic peaks are evident between ~2.4–2.8 V ($Cu^0 \rightarrow Cu^+$) and at 3.7 V ($Cu^+ \rightarrow Cu^{2+}$). The formation of soluble ionic species enables diffusion through the separator, which may lead to crossover (i.e., cathode contaminant deposition at the anode) and may also induce a hard internal short through the formation of lithium dendrites. In full cells, it is anticipated that ionized Cu^+ and Cu^{2+} migrating to the anode may incorporated into the SEI. This is anticipated to induce a deleterious effect on SEI growth, leading to both excessive Li consumption and the formation of a resistive passivating layer that inhibits Li-ion transport. As such, an extreme reduction in cyclable capacity is observed for full cells containing metallic Cu^0 (Fig. A7(e)).

While the erratic differential capacity signature observed for Cu^0-contaminated half cells in Fig. 5(e) suggests an internal short, an interesting capacity recovery is observed during subsequent cycling (Fig. A6(e)). Following what appears to be a short during initial formation cycles, half cells containing Cu^0 in the cathode then show only a slightly

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depressed capacity relative to the baseline throughout subsequent cycling. The capacity drop towards the end of cycling implies a restriction on the functionally infinite Li reservoir, perhaps suggesting an early deposition and passivation of Cu on the Li surface. This would lead to regions of "dead" Li, which could no longer participate in electrochemical reaction. The observed capacity recovery in half cells containing Cu^0 impurity at the cathode runs counter to previous reports on the impacts of dissolved copper in Li-ion half cells [11,25], which have suggested that ionic Cu^+/Cu^{2+} is deleterious to half cell operation.

3.3.5. Si^0-contaminated cathode

As shown in Fig. 5(f), cathodes containing Si^0 impurity show evidence of significant first-cycle reactivity at high potentials (above 4.0 V) in half-cell format. This reactivity is largely limited to the first formation cycle, with half-cell differential capacity signatures stabilizing after the first cycle. This behavior may be attributed to Si^0 reactivity with the electrolyte, and in particular, the evolution of oxygen that reacts with Si to form silicon-oxide species (e.g., SiO_2), which may subsequently react with lithium to form electrochemically stable Li_2SiO_3. In the presence of an infinite Li^0 reservoir (i.e., half cell), this loss of lithium does not significantly impact capacity performance in cells containing Si^0-contaminated cathodes (Fig. A6(f)). However, in full cells, a significant decline in capacity is observed (Fig. A7(f)). Notably, the capacity of cells with Si^0 impurities at the cathode remains stable throughout cycling, with the entirety of the capacity loss occurring within the first 10 cycles. This finite reactivity may be linked to the initial quantity of impurity present, and suggests that cathode-sited Si^0 impurity will continue to react with electrolyte-derived oxygen and Li until the Si_0 has been fully passivated to Li_2SiO_3.

The thermal signatures for full cells with Si^0 contaminant at the cathode (Fig. 7(e)) align well with the proposed mode of electrochemical reactivity; namely, the strong exothermic reactivity above 3.5 V may be tied to the formation of SiO_2, and then subsequently Li_2SiO_3.

Differential capacity analysis of full cells with Si^0-contaminated cathodes (Fig. 6(f)) suggests a shift to higher potentials of the two dominant oxidative redox peaks with formation cycling (i.e., from cycles 1 to 4) and a concurrent increase in hysteresis and decline in the intensity of both dominant redox peaks. This implies a growth in impedance, ostensibly due to the deposition of evolved Li_2SiO_3 on the anode.

3.3.6. Summary of cathode contaminant behavior

In terms of practical cell performance, Cu^0 and Si^0 impurities at the cathode most significantly impact long-term reversible cycling capacity. For all other contaminants, effective passivation occurs during the first cycle, leading to a gravimetric capacity penalty but minimal additional impact on cell performance. Notably, none of the cathode contaminants are found induce a hard internal short (e.g., through separator penetration) under the studied conditions. Crossover is anticipated in the case of the Cu^0 impurity at the cathode, whereas the mode of reactivity for the Si^0 impurity is more likely linked to a continuous passivation over numerous cycles. Thermal signatures neatly complement the observed electrochemical reactivity in most cases; however, for the Fe^0 contaminant, additional thermal signals are observed that cannot readily be paired with electrochemical signatures. This highlights the need for complementary analysis and may also direct further study into the mechanism of Fe^0 reactivity under cathodic conditions.

CRediT authorship contribution statement

Kae E. Fink: Writing – original draft, Revised Draft, Responses to Reviewers, Formal analysis, Visualization. **Bryant J. Polzin:** Conceptualization, Methodology, Writing – original draft, Funding acquisition, Project administration. **John T. Vaughey:** Formal analysis. **Joshua J. Major:** Investigation, Formal analysis, Software. **Alison R. Dunlop:** Investigation, Data curation, Visualization. **Stephen E. Trask:** Investigation. **Gerald T. Jeka:** Investigation. **Jeffrey S. Spangenberger:** Supervision. **Matthew A. Keyser:** Conceptualization, Methodology, Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2021.230760.

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